



Ai2th

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Linxr | Part 1 — The Idea and Architecture

#android #linux #qemu #architecture

Linxr — Alpine Linux on Android (12 Part Series)

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The Idea and Architecture

Part 1 of 4 — building Linxr, a single APK that runs Alpine Linux on non-rooted Android.

The Constraint

Android is a locked-down Linux system. For regular (non-root) apps:

- `execve()` from app storage is blocked by SELinux W^X policy
- Native Linux tools are unavailable without Termux

- The kernel disables `CONFIG_USER_NS` — Docker rootless mode is impossible
- No way to load kernel modules or create network namespaces

A QEMU virtual machine sidesteps all of this. The VM runs its own kernel with its own namespace — completely isolated from Android's restrictions.

Why Alpine?

Alpine Linux is the right guest OS for a mobile VM:

Property	Alpine	Debian/Ubuntu
Compressed disk image	~35 MB	~300 MB+
Boot time	~5 s	~30+ s
RAM at idle	~30 MB	~100 MB+
Package manager	apk	apt
Init system	OpenRC	systemd

Small footprint, fast boot, full POSIX shell — ideal for embedding in an APK.

Architecture

```
Android OS (non-rooted)
├── APK (Flutter + Kotlin)
│   ├── VmManager - asset extraction, QEMU lifecycle
│   ├── VmService - ForegroundService (keeps QEMU alive)
│   └── Flutter UI - Home, Terminal, About tabs
│       └── TerminalScreen (dartssh2 + xterm)
└── QEMU process (libqemu.so)
    ├── Alpine Linux 3.19 VM
    └── OpenRC init → sshd on :22
        ↑
```

```
SLIRP hostfwd tcp::2222-:22
↑
SSH on 127.0.0.1:2222 (from Android app)
```

Disk Layout

Linixr uses QCOW2 copy-on-write layering:

```
base.qcow2 (read-only - Alpine rootfs, openssh baked in)
└─ user.qcow2 (writable overlay - your data lives here)
```

- `base.qcow2` ships compressed in the APK assets. Extracted once on first install.
- `user.qcow2` is created fresh on first install. Persists across VM restarts — installed packages and files survive.

The overlay is only recreated when `base.qcow2` changes (asset version marker bumps on app update).

SLIRP Networking

QEMU SLIRP requires no root and no kernel modules:

```
Android app
└─ TCP to 127.0.0.1:2222
    └─ QEMU hostfwd: tcp::2222-:22
        └─ Alpine VM eth0 (10.0.2.15):22
            └─ sshd
```

Inside the VM:

```
eth0:      10.0.2.15/24
gateway: 10.0.2.2
DNS:       10.0.2.3 (SLIRP built-in resolver)
```

The VM has full outbound internet access. `apk add`, `curl`, `git` all work normally.

Token-Free Design

Pockr uses a UUID token injected via QEMU kernel cmdline to authenticate its HTTP API. Linxr doesn't need that — SSH handles authentication natively:

- Username: `root`
- Password: `alpine` (baked into the base image)

No tokens, no custom API — standard SSH.

Next: [Part 2 — Shipping QEMU in an APK](#)

GitHub: github.com/AI2TH/Linxr

Linxr Series — Alpine Linux on Android

Linxr = *Linux* + *r*. A single Android APK that runs a full Alpine Linux shell on any Android phone — no root, no Termux, no PC required.

#	Post	Topic
	Intro	What is Linxr? Start here
1	Part 1	The Idea and Architecture
2	Part 2	Shipping QEMU in an APK
3	Part 3	SSH Terminal in Flutter
4	Part 4	Test Results

GitHub: github.com/AI2TH/Linxr

Website: ai2th.github.io

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Sentry PROMOTED



The screenshot displays the Sentry web interface for a React Native application. The top navigation bar includes the Sentry logo and a 'PROMOTED' badge. The main content area is divided into two sections: 'Breadcrumbs' and 'Logs'. The 'Breadcrumbs' section shows a sequence of events leading up to an error, with the first two items highlighted by red boxes. The first breadcrumb is an 'Exception' with the message 'Error: API Error: 500'. The second breadcrumb is an 'api-error' with the message 'POST /api/votes failed with status 500' and a detailed error object. The 'Logs' section shows a series of app lifecycle events, including 'App entered background', 'App state changed', and 'App became active'.

React Native logs that actually help you debug production issues

Improve your mobile debugging workflow with structured logs in React Native that provide actual context for production crashes.

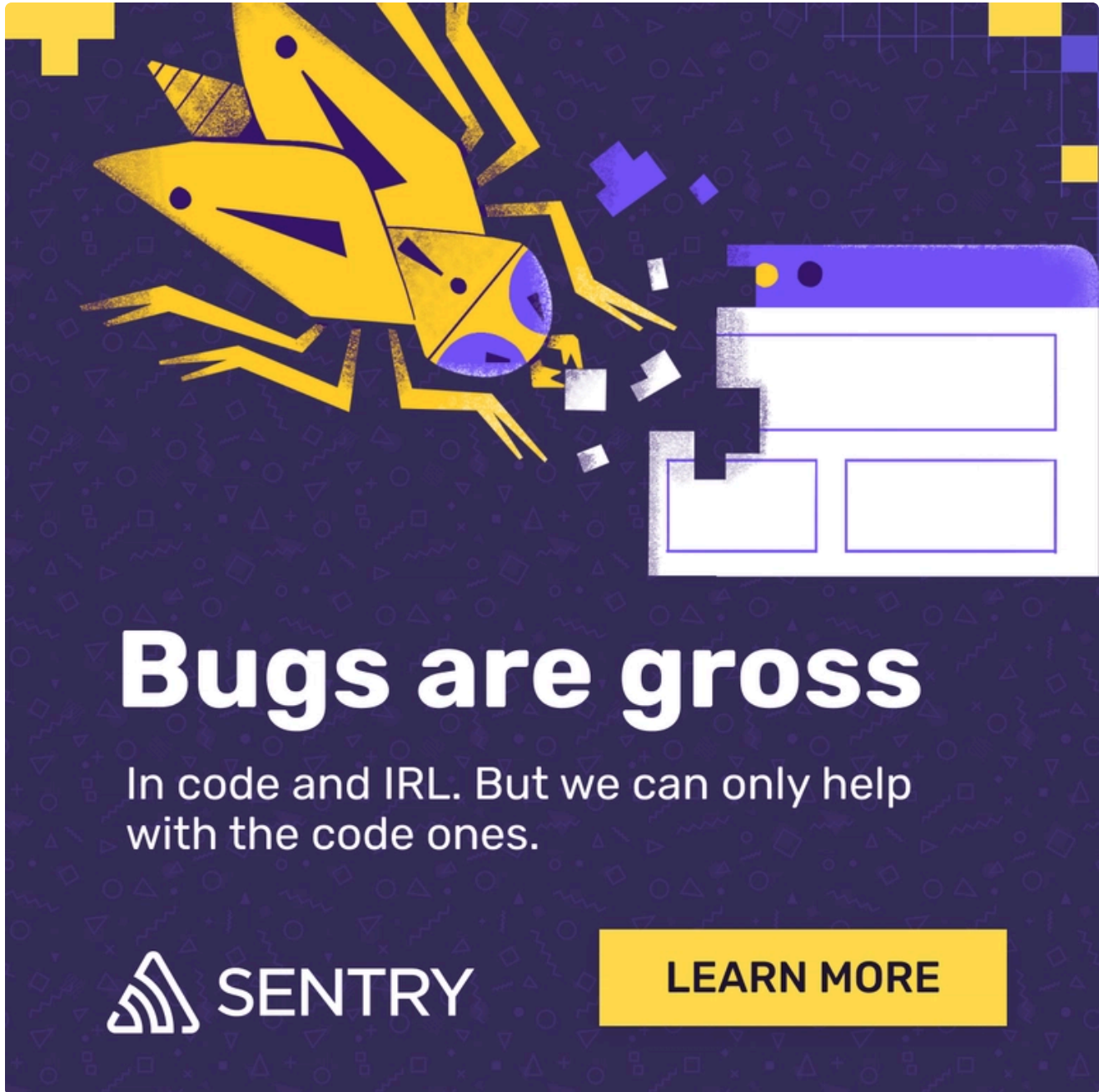
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 Sentry PROMOTED

...

The advertisement has a dark blue background with a pattern of small, light blue geometric shapes. At the top, a stylized yellow bug is shown from a top-down perspective, appearing to crawl towards the right. The bug has a segmented body, six legs, and a head with two antennae. To the right of the bug is a white rectangular box representing a code editor, with a yellow border and a small yellow circle in the top-left corner. Below the bug and the code editor, the text "Bugs are gross" is written in a large, bold, white sans-serif font. Underneath this, in a smaller white font, is the text "In code and IRL. But we can only help with the code ones." At the bottom left, the Sentry logo (a stylized 'S' made of three curved lines) is followed by the word "SENTRY" in a bold, white, sans-serif font. At the bottom right, there is a yellow rectangular button with the text "LEARN MORE" in a bold, black, sans-serif font.

Bugs are gross

In code and IRL. But we can only help with the code ones.

 SENTRY

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See why 4M developers consider Sentry, “not bad.”

Fixing code doesn't have to be the worst part of your day. Learn how Sentry can help.

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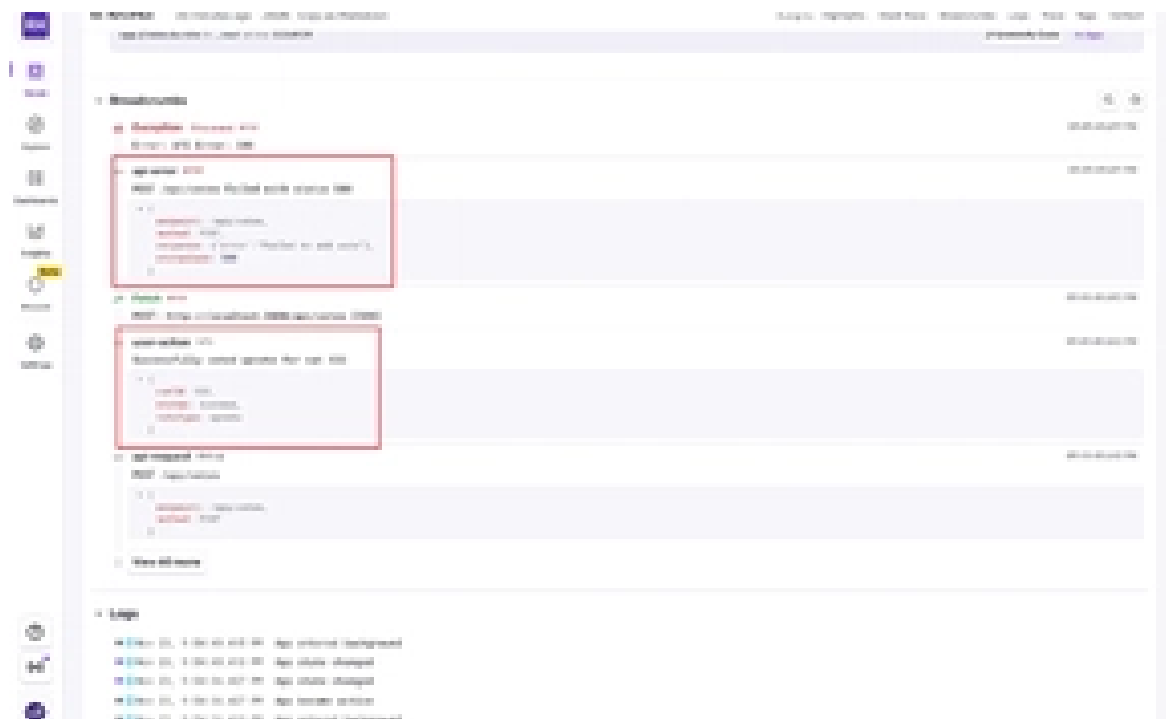
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Sentry

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